1. **Which of the following is an example of a supervised learning problem?**
   A) Clustering customers based on their purchase history
   B) Identifying spam emails based on labeled training data
   C) Segmenting an image without prior knowledge
   D) Finding patterns in an unlabeled dataset

2. **What is the purpose of a cost function in machine learning?**
   A) To compute the accuracy of the model
   B) To evaluate how well the model's predictions match the actual values
   C) To visualize the dataset
   D) To standardize the input features

3. **Which of the following is NOT a type of neural network?**
   A) Convolutional Neural Network (CNN)
   B) Recurrent Neural Network (RNN)
   C) Decision Tree Neural Network (DTNN)
   D) Transformer

4. **In machine learning, what does the term "bias" refer to?**
   A) The difference between training and test accuracy
   B) The model's ability to generalize to new data
   C) The error due to overly simplistic assumptions in the model
   D) The tendency of a model to memorize training data

5. **Which of the following algorithms is best suited for detecting anomalies in data?**
   A) K-Means Clustering
   B) Isolation Forest
   C) Naïve Bayes
   D) Random Forest

6. **What is the primary objective of Principal Component Analysis (PCA)?**
   A) To increase the number of features in a dataset
   B) To transform categorical variables into numerical values
   C) To reduce the dimensionality of the dataset while preserving variance
   D) To cluster data into distinct groups

7. **Which evaluation metric is used for measuring the performance of a classification model?**

   A) Mean Absolute Error

   B) R-squared

   C) F1-score

   D) Sum of Squared Errors

8. **What is the key advantage of using a Support Vector Machine (SVM)?**
   A) It is easy to interpret like a decision tree
   B) It is efficient on large datasets
   C) It is effective when there is a clear margin of separation between classes
   D) It requires very little computational power
9. **Which technique is commonly used to handle imbalanced datasets?**
   A) Increasing the batch size
   B) Oversampling the minority class
   C) Reducing the number of features
   D) Applying dropout

10. **What does the term "hyperparameter tuning" refer to?**
   A) Adjusting the weights of a neural network during training
   B) Finding the best set of parameters that control the learning process
   C) Evaluating a model on a test dataset
   D) Optimizing the dataset size for training

11. **Which of the following methods can be used to prevent overfitting?**
   A) Increasing the number of hidden layers in a neural network
   B) Using more epochs for training
   C) Applying L1 or L2 regularization
   D) Decreasing the learning rate

12. **Which machine learning technique is used to improve model performance by assigning different weights to different classifiers?**

   A) Bagging

   B) Boosting

   C) Clustering

   D) Normalization

13. **In machine learning, what does an epoch refer to?**

a) A single pass through the entire training dataset ✅
b) The number of layers in a neural network
c) The total number of test samples
d) The process of feature selection

14. **What is the main purpose of dropout in neural networks?**
    A) To speed up training
    B) To prevent overfitting by randomly turning off neurons during training
    C) To increase model complexity
    D) To reduce bias
    Answer: B) To prevent overfitting by randomly turning off neurons during training

15. **What is a key benefit of using ensemble learning?**
    A) It simplifies the model
    B) It reduces computational complexity
    C) It combines multiple models to improve overall performance
    D) It ensures 100% accuracy
    Answer: C) It combines multiple models to improve overall performance